

Day 15

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B.P.S.C

Q Study following Table Carefully
and answer question given below Table

Three Companies X, Y, Z making identical Soap

$$\text{Total Sale} = \text{Selling Price} \times \text{Unit Sold}$$

Sol

$$\text{Unit Sold} = \frac{\text{Total Sales}}{\text{Selling price}}$$

Year	X		Y		Z	
	Unit Price (₹/Unit)	Sales (₹/Lakh)	Unit price (₹/Unit)	Sales (₹/Lakh)	Unit price (₹/Unit)	Sales (₹/Lakh)
2011	5	3	4	4.4	7	3.5
2012	6	3	6	4.2	6	3.6
2013	7	3.5	5	4.5	8	4
2014	8	4	7	4.9	5	5
2015	9	3.6	6	4.2	6	4.2

a) What was average price of Soap in year 2013 (₹/unit)

Sol

From above given Table

Price of Soap in year 2013
of Three Companies X, Y, Z

$$X \rightarrow 7 \text{ (₹/unit)}$$

$$Y \rightarrow 5 \text{ (₹/unit)}$$

$$Z \rightarrow 8 \text{ (₹/unit)}$$

$$\text{Average Price in 2013} = \frac{\text{Total Price of Soap}}{\text{Total Number of Soap}}$$

$$= \frac{7+5+8}{3} = \frac{20}{3} = 6.67 \text{ ₹/unit}$$

$$\therefore \boxed{\text{Avg Price in 2013} = 6.6 \text{ ₹/unit}}$$

b) What was percentage decrease
in unit sold for X in 2012 over 2011?

Sol

In 2012 year

Unit Price of X Soap = ₹ (2/unit)

Total Sales of Soap = ₹ 3,00,000

$$\text{No of Unit sold} = \frac{3,00,000}{2} \times \frac{(2/\text{unit})}{(2)}$$

$$= 50,000 \text{ Unit}$$

Similarly

In 2011 year

Unit price of X Soap = ₹ (5/unit)

Total Sales of Soap = ₹ (3,00,000)

$$\text{No of Unit sold} = \frac{3,00,000}{5} = 60,000 \text{ Unit}$$

$$\therefore \text{Percentage decrease} = \frac{10,000}{60,000} \times 100$$

in 2012 over 2011

$$\text{required \% decrease} = \frac{100}{6} = \frac{50}{3} = 16.67 \%$$

C What was percentage share of
Z in total unit sold in 2013?

Sol

In 2013 year

$$\begin{aligned}\text{Total unit sold by X} &= \frac{\text{Total Sales}}{\text{Unit Price}} \\ &= \frac{350000}{7} \text{ ₹/unit}\end{aligned}$$

Similarly,

$$X = 50,000 \text{ Unit}$$

$$\text{Total Unit Sold by Y} = \frac{450000}{5} = 90,000 \text{ Unit}$$

$$\text{Total Unit Sold by Z} = \frac{400000}{8} = 50,000 \text{ Unit}$$

$$\begin{aligned}\text{Total Unit Sold by X, Y, Z} &= 50,000 + 90,000 + 50,000 \\ &= 1,90,000 \text{ Unit}\end{aligned}$$

$$\text{Percentage Share of Z in Total unit Sold} = \frac{50,000}{1,90,000} \times 100$$

$$\text{required \%} = \frac{500}{19} = 26.31 \%$$

Row

Q) In which year Y Sold maximum Number of Unit?

Sol

Soap Sold by Y in different year

$$2011 \rightarrow \frac{4,40,000}{4} = 1,10,000 \text{ Unit}$$

$$2012 \rightarrow \frac{4,20,000}{6} = 70,000 \text{ Unit}$$

$$2013 \rightarrow \frac{4,50,000}{5} = 90,000 \text{ Unit}$$

$$2014 \rightarrow \frac{4,90,000}{7} = 70,000 \text{ Unit}$$

$$2015 \rightarrow \frac{4,20,000}{6} = 70,000 \text{ Unit}$$

From above data it is clear that

Maximum Number of Unit Sold

by Y is in 2011 year i.e.

1,10,000 Unit

e

Q From given Table

Maximum price per unit = 9 ₹/unit
is in 2015 by X Company

Minimum price per unit = 4 ₹/unit
is in 2011 by Y Company

$$\begin{aligned}\text{Required percentage} &= \frac{(9-4) \text{ ₹/unit} \times 100}{4 \text{ ₹/unit}} \\ &= \frac{5}{4} \times 100 = 125\%\end{aligned}$$

$$\boxed{\% = 125\%}$$